



APEXA RAMI

Generative AI & Software Engineer

✉ apexajrami@gmail.com ☎ +49 15565865428

📍 Ziegetsdorferstr 78, 93051, Regensburg 🔗 [linkedin.com/in/apexa-rami](https://www.linkedin.com/in/apexa-rami)

🔗 apexarami.github.io

SUMMARY

Software developer with a strong foundation in Python, backend development, and cloud technologies. Experienced in building scalable systems, REST APIs, and working with AWS and CI/CD pipelines. Recently focused on machine learning, Generative AI, and MLOps, with hands-on experience in developing LLM-based workflows and integrating AI into real-world applications.

WORK EXPERIENCE

10/2025 – 03/2026
Belgium

Research Intern – Generative AI (LLMs), *Materialise* 🌐

- Worked in the Central Research and Technology unit, contributing to the development of Generative AI solutions using Large Language Models for enterprise use cases.
- Designed and implemented a Deep Research system capable of multi-step reasoning, iterative retrieval, and structured response generation, along with web search-enabled AI workflows to improve real-time information access.
- Developed backend services using FastAPI and integrated LLM APIs to support scalable AI-driven functionalities.
- Deployed and managed services on AWS and contributed to Azure-based MLOps pipelines for model integration and workflow automation.
- Additionally, worked on improving system architecture and enhancing the codebase to increase scalability, performance, and overall research efficiency while collaborating with cross-functional teams.

11/2022 – 11/2023
Ahmedabad, India

Software Developer, *Syndell technologies* 🌐

- Developed web applications and RESTful APIs using PHP, Node.js, and Java.
- Built responsive front-ends with HTML, CSS, JavaScript, and Bootstrap.
- Delivered e-commerce platforms with payment systems, order tracking, and supplier API integrations.
- Managed databases (PostgreSQL, MongoDB) and deployed applications via Docker, AWS/GCP, and CI/CD pipelines.
- Contributed in Agile teams through code reviews, sprint planning, and documentation.

03/2021 – 12/2021
Bhavnagar, India

Intern Web-designer & SEO expert, *IT-HUB Software development*

- Designed responsive websites, prioritizing UX.
- Implemented SEO strategies, boosting traffic.
- Conducted keyword research and content optimization.
- Used Google Analytics for performance tracking.
- Collaborated with clients to deliver solutions.
- Provided ongoing maintenance and support.

02/2020 – 01/2021
Vadodara, India

Intern in Home Automation Product manufacturing, *Varni Digital*

- Assisted in the development and testing of IoT-based home automation products using Python.
- Conducted market research to guide product improvements and collaborated with teams to address technical challenges.
- Participated in product demonstrations and feedback sessions, contributing to documentation and reporting.

- Gained practical skills in IoT, Python, and industry knowledge.

EDUCATION

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|---------------------------------|--|
| 03/2024 – Present
Regensburg | Master of Engineering in Electrical and Microsystem,
<i>Ostbayerische Technische Hochschule (OTH) Regensburg</i> <ul style="list-style-type: none"> • Major Subjects: Electronics Production Engineering, Introduction to AI & Machine Learning, Advanced Engineering Mathematics, Deep learning, OOP with Python, Applied Optics, LED Technology, Applied Optoelectronics, Project Management, Advanced Semiconductor Technology. |
| 05/2018 – 05/2022 | Bachelor of Engineering in Electronics and communications,
<i>Gujarat Technological University</i> <ul style="list-style-type: none"> • Specialized in Artificial Intelligence, IoT, and VLSI Design. • Gained practical experience in AI algorithms, IoT protocols, and VLSI simulations. • Strong foundation in C programming for embedded systems and algorithm development. • Completed projects integrating AI and IoT for smart system applications. |
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PROJECTS

GenAI Document Deep Research Assistant [🔗](#)

Goal: Develop a local research assistant for extracting reliable answers from technical PDF documents using retrieval based search and local language model support.

Description:

- Built a FastAPI and Streamlit application for uploading PDFs and querying document content.
- Implemented deep research mode with query expansion, evidence retrieval, page citations, and simplified answer generation.
- Added optional Ollama integration to generate answers locally without external API dependency.

Tools: Python, FastAPI, Streamlit, Ollama, Llama 3.2, scikit learn, pypdf, PyMuPDF, Docker, GitHub Actions

Smart Grid Load Forecasting with MLOps [🔗](#), Case-study

Goal: Build a machine learning based forecasting workflow for estimating electricity demand from weather, time, and grid operation features.

Description:

- Developed a FastAPI service and Streamlit dashboard for smart grid load prediction.
- Created a training pipeline with synthetic grid data, model evaluation, and saved inference pipeline.
- Added Docker support, API tests, and GitHub Actions CI for reproducible execution.

Tools: Python, FastAPI, Streamlit, scikit learn, Pandas, NumPy, joblib, Docker, Pytest, GitHub Actions

Industrial IoT Anomaly Detection Pipeline [🔗](#)

Goal: Develop a real time monitoring pipeline for detecting abnormal machine behavior from industrial sensor streams.

Description:

- Built an MQTT based sensor pipeline for streaming machine temperature, vibration, current, pressure, and speed values.
- Implemented anomaly detection using rule based checks and an Isolation Forest model for machine health scoring.
- Added Docker Compose setup with Mosquitto, InfluxDB, and Grafana for time series monitoring and dashboard visualization.

Tools: Python, MQTT, Paho MQTT, scikit learn, Pandas, InfluxDB, Grafana, Docker Compose, Pytest, GitHub Actions

Vision Quality Gate with OpenCV and Gradio [🔗](#)

Goal: Develop a computer vision based quality gate for checking image usability and visible surface irregularities in inspection images.

Description:

- Built an interactive Gradio demo for uploading inspection images and visualizing quality gate results.
- Implemented OpenCV based checks for blur, glare, fog or smoke like degradation, and surface irregularities.

- Generated annotated outputs with defect mask, inspection score, decision label, and recommended action.

Tools: Python, OpenCV, Gradio, NumPy, Pillow, Pytest, GitHub Actions

Inventory Management System, Client Project

- Technologies: Python, Django, PostgreSQL, REST API, AWS
- Built a web-based inventory management system for real-time stock control.
- Integrated supplier APIs and implemented role-based authentication.
- Optimized database performance and deployed the application on AWS.

Website Performance Optimization (PageSpeed Improvement Project), Client Project

- Technologies: React, Next.js, Webpack, Lighthouse, Google Cloud
- Boosted Google PageSpeed score from 61 to 95 by optimizing image delivery, minifying JS/CSS, and implementing server-side rendering.
- Reduced bounce rate by 30% and increased average session time through faster load times.
- Deployed optimized assets via CDN for global performance consistency.

CI/CD Pipeline Automation (DevOps Efficiency Initiative), Client Project

- Technologies: GitLab CI/CD, Docker, Kubernetes, AWS
- Automated build, test, and deployment workflows to reduce release time by 60%.
- Integrated static code analysis and unit tests into the pipeline for quality assurance.
- Enabled rollbacks and blue-green deployments to minimize downtime.

SKILLS

GenAI and Agentic AI: — Agentic AI architecture, Deep Research agents, multi agent workflows, tool calling, function calling, planner executor patterns, reflection loops, LangChain, LangGraph, Azure OpenAI, OpenAI Responses API, LibreChat, MCP servers, Ollama

Backend and Cloud: — Node.js, MongoDB, Python, FastAPI, REST APIs, Pydantic, Docker, Docker Compose, AWS ECS, AWS ECR, Azure ML, GitLab CI/CD, GitHub Actions

Cloud & MLOps — AWS (ECS), Azure OpenAI, Azure MLOps, Docker, CI/CD (GitLab, YAML)

Simulation — Siemens Tecnomatix Plant Simulation (Python integration)

RAG and Search: — RAG, embeddings, vector search, hybrid search, document chunking, reranking, Cohere, SearXNG, Firecrawl, citation based responses, document Q&A

Databases and Data: — PostgreSQL, MongoDB, SQLite, InfluxDB, Pandas, NumPy, data pipelines, time series data

ML and Computer Vision: — scikit learn, anomaly detection, forecasting, classification, regression, OpenCV, image processing, image quality inspection

LANGUAGES

English — Native/Bilingual
IELTS

German — Conversational